	GLOBAL ECONOMICS **Grade: 10 - Learning Evidence 3** **DECISION-MAKING UNDER UNCERTAINTY** **Version 2** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.	**Assessment criteria:** **C8**: Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations. **C9**: Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative. **C10**: Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

Name: _____ **Grade:** _____ **Date:** _____

1. Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

Question	**C8**	**C9**
1	3	–
2	3	–
3	–	2
4	–	2
5	–	2
Total	**6**	**6**

2. 1. Camera Rental Plans [C8-3]

A camera rental studio selects its inventory under uncertain demand. Profits are in thousand USD.

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.6	0.2
Essential (A_1)	100	90	70
Shared (A_2)	140	150	40
Extensive (A_3)	240	48	-40
Modular (A_4)	170	130	-20

Decision statistics (Regret = maximum regret):

Alt.	Max	Min	Regret	EMV
A_1	100	70	140	88
A_2	150	40	100	126
A_3	240	-40	110	68.8
A_4	170	-20	90	108

(a) Using the original camera rental payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A subsidy s (integer, $0 \leq s \leq 100$) increases the low-demand payoffs of A_3 and A_4 by s. An alternative is eligible only if all its payoffs are nonnegative. Determine each camera rental plan's eligibility based on the value of s. **[0.5]**

(c) Among eligible camera rental plans, find the smallest allowed subsidy s at which the Maximin recommendation changes from A_1 to A_4 alone. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $0 \leq s \leq 100$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

3. 2. Outdoor Cinema Refreshments [C8-3]

A cinema manager selects a refreshment kiosk under uncertain audience size. Profits are in hundred USD.

Alternative	High	Low
$P(S_j)$	0.4	0.6
Portable (C_1)	96	64
Full-service (C_2)	156	20
Deluxe (C_3)	180	-10
Fixed (C_4)	140	50

Decision statistics (Regret = maximum regret):

Alternative	Max	Min	Regret	EMV
C_1	96	64	84	76.8
C_2	156	20	44	74.4
C_3	180	-10	74	66
C_4	140	50	40	86

(a) Using the original refreshment kiosk payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A sponsor replaces C_1's large-audience payoff of 96 with g (integer, $96 \leq g \leq 160$). An alternative is eligible only if all its payoffs are at least 40. Determine each refreshment kiosk's eligibility based on the value of g. **[0.5]**

(c) Among eligible refreshment kiosks, find the smallest allowed sponsored payoff g at which the Minimax Regret recommendation changes from C_4 to C_1 alone. Recalculate regret benchmarks using only eligible kiosks. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $96 \leq g \leq 160$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

4. 3. Garden Centre Stock [C9-2]

A garden centre selects a seedling stock plan D_1–D_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
D_1	140	60	60	118
D_2	160	40	40	132
D_3	120	80	80	106
D_4	180	20	60	132
D_5	100	60	100	86
D_6	200	0	80	132

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

5. 4. Climbing Gym Expansion [C9-2]

A climbing gym selects an expansion plan F_1–F_6. Net profits are in thousand USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
F_1	190	50	50	137
F_2	150	90	90	128
F_3	170	70	70	146
F_4	240	0	90	122
F_5	130	80	110	110
F_6	210	30	60	129

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

6. 5. Art Materials Orders [C9-2]

An art materials distributor selects an order plan H_1–H_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
H_1	150	90	90	128
H_2	170	70	70	144
H_3	210	30	60	129
H_4	130	80	110	110
H_5	240	0	90	122
H_6	190	50	50	137

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

7. C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.